

LIR Source

```
require cap(relay1)
require cap(water_sensor)
set relay1 = 1
wait 500ms
read water_sensor
halt
```

*LLM-generated
running example*

Compiler

lower

```
cap(5)    → GTWAY 5
cap(1)    → GTWAY 1
set 1     → LIT 1; IOW 5
wait 500  → WAIT 500
read s    → IOR 1
halt     → HALT
```

encode

*O(n) deterministic
unique output*

Bytecode

```
GTWAY 5
GTWAY 1
LIT 1
IOW 5
WAIT 500
IOR 1
HALT
```

*12 bytes
varint encoded*